

2. Enjoy meat in moderate amounts

Environment: Meat production significantly affects climate change, as the “impacts of the lowest-impact animal products typically exceed those of vegetable substitutes” (Poore and Nemecek 2018). At the highest end of CO₂-eq emissions is ruminant meat, i.e. beef and lamb, which can be almost 10 times higher than poultry, eggs, rabbit and most fish, and 4 to 5 times higher than pork (Clune et al. 2017, Poore and Nemecek 2018). In terms of pressure on freshwater resources, meals that include beef have 3 times the WF as those that include other meat protein sources (pork, chicken) or legumes, and twice the WF of eggs (Mekonnen, M.M. and Hoekstra, A.Y., 2010). The big WF impact of beef is mainly related to the production of animal feed over the lifetime of the animal (Hoekstra 2015). A WF reduction of up to 35% (average per capita per day) could be achieved by adopting the common guidelines for healthy diets (Vanham et al. 2018). A good compromise for protein intake is represented by fish, both in terms of CO₂ equivalent emissions and water saving (Mekonnen, M.M. and Hoekstra, A.Y., 2010, Clune et al. 2017). A full shift to a pescatarian (or vegetarian) diet could reduce the WF per capita within the range 33-55% (Vanham et al. 2018).

Health: Following reclassification in 2015 by the World Health Organization’s International Agency for Research on Cancer, processed meats (such as hot dogs, ham, sausages, corned beef, canned meat, and meat-based sauces) eaten daily may increase the risk of colorectal cancer and stomach cancer. A minimum intake of these types of food that guarantees complete safety cannot be defined. Similarly, also red meat consumption (i.e. all muscle meat, such as beef, veal, pork, and lamb) is associated with an increased risk of colorectal, pancreatic, and prostate cancers (IARC, 2015), cardiovascular disease (Abete et al. 2014), and risk of stroke and type 2 diabetes (Chen et al., 2013, Feskens et al., 2012). Such risks can be significantly diminished by substituting other protein sources for red meat (Pan et al., 2012).

According to the EAT-Lancet commission (Willett et al. 2019): “Because intake of red meat is not essential and appears to be linearly related to total mortality and risks of other health outcomes in populations that have consumed it for many years, optimal intake might be 0 g/day, especially if replaced by plant sources of protein. Because data on risk of low intakes of red meat are imprecise, we have concluded that an intake of 0 g/day to about 28 g/day of red meat is desirable and have used a midpoint of 14 g/day for the reference diet. Since consumption of poultry has been associated with better health outcomes than red meat, we have concluded that the optimum consumption of poultry is 0 g/day to about 58 g/day and have used a midpoint of 29 g/day for the reference diet.”

3. Enjoy dairy products in moderation

Environment: Plant-based milk, except for rice milk, has a lower Carbon Footprint (CF) than dairy milk (Clune et al. 2017). The difference is even more significant for the Water Footprint (WF of soy milk is 296.9 L/kg vs cow's milk 1050 L/kg, Ercin et al. 2011). Yogurt has a value similar to milk, while cheese - due to the high concentration of milk used for its production - can have a CF and WF up to 6 times higher than the original milk. When compared with other protein sources, cheese has twice the WF of soy-based proteins, and both the CF and WF of cheese can be higher than those of many meat sources, including chicken, pork and fish (Clune et al. 2017, Mekonnen, M.M. and Hoekstra, A.Y., 2010).

Health: High intake of dairy products, at least three servings per day, has been widely promoted in Western countries for bone health and fracture prevention, primarily because of their high calcium content. However, the optimum calcium intake remains uncertain (Willett et al., 2019), while there is evidence that the adverse effect of protein of animal origin might outweigh the positive effect of calcium intake on the calcium balance (FAO and WHO, 2002). Animal-based dairy fat like butter conveys a higher risk of cardiovascular disease than unsaturated plant oils, which therefore should be preferred whenever possible for cooking and food preparation (EAT-Lancet Commission, Willett et al., 2019).

4. Avoid too much food

Environment: Every kilogram of food carries the embedded environmental impact generated during its production, processing, transportation, and packaging (Willett et al. 2019, Ercin et al. 2011, Clune et al. 2017, Vanham and Bigoglio, 2013). To reduce pressure on the environment, we need to not only balance the composition of our diet, but also to reduce the portions of food we consume. Every time we throw food into the waste bin, we are basically wasting resources and generating unnecessary environmental pressures. The wasted environmental value is higher for meat and animal-based food (Clune et al. 2017, Willett et al. 2019). A dietary shift combined with reducing our food waste might help us to reach important sustainability targets in terms of climate change and water use (Willett et al. 2019). In Spain, for example, the WF of food waste accounts for 2095 hm³, equivalent to 131 litres per person per day (Blas et al. 2018).

Health: Energy intake (calories) should be in balance with energy expenditure (WHO, 2018). Excess energy intake (i.e. energy-dense unhealthy dietary patterns) has been identified as the most significant dietary factor in relation to weight gain and the development of obesity (Roberts et al., 2002).